

Wang’s Five Laws: Spectral Signals Inside Large Language Model Attention Weights

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Reasoning capability in large language models (LLMs) is typically assessed through benchmark evaluation, which is costly, prompt-sensitive, and requires full inference. We ask whether reasoning capacity can be inferred directly from static attention projection weights, without running any inference. By applying singular value decomposition (SVD) to the Query (Q), Key (K), and Value (V) weight matrices across five open-source LLM families, we identify five universal cross-model regularities that we formalize as *Wang’s Five Laws*.

The **First Law** (Spectral Linear Alignment) states that the Pearson correlation between Q and K singular-value spectra converges toward $r_{\text{Wang}} = 1$, observed in practice at 0.94–0.99. The **Second Law** (Spectral Shape Fidelity) states that the normalized spectral mismatch (SSR) decreases systematically in deep layers and is further reduced by reinforcement-learning (RL) fine-tuning, with RL-tuned models showing up to 3.92% lower deep-layer SSR than their base counterparts. The **Third Law** (Precision-Depth-Logic Criterion) derives an analytic upper bound on trainable depth as the minimum of three independent limits, explaining why models deeper than 40–80 layers universally adopt BF16 or mixed precision. The **Fourth Law** (Global Output-Subspace Decoupling) states that the mean cosine similarity between matched left singular vectors satisfies $\overline{\cos}(U_Q, U_K) \sim 1/\sqrt{d_h}$ (random orthogonality), while $\overline{\cos}(U_Q, U_V)$ and $\overline{\cos}(U_K, U_V)$ are *systematically below* the random baseline (*super-orthogonality*), guaranteeing functional isolation between the retrieval path and the content-readout path. The **Fifth Law** (Global Random Orthogonality of Input Subspaces) states that all pairwise cosine similarities between right singular vectors of Q, K, V are close to the random baseline $1/\sqrt{d_{\text{model}}}$, confirming that the three projections independently and freely select sensitive directions from the input embedding space.

Together, the five laws reveal a *universal geometric fixed point* reached at the end of pretraining: singular-value spectra are fully aligned (Σ -space), output subspaces are functionally decoupled with super-orthogonality between V and Q/K (U -space), and input subspaces are globally random-orthogonal (V -space). All findings are reproducible from publicly available model weights using the scripts at <https://github.com/emis-framework/math-under-llm>.

I. INTRODUCTION

A. Motivation

Benchmark-based evaluation of LLM reasoning is expensive and prompt-sensitive [1]. Running a full evaluation suite on a single checkpoint may require thousands of GPU-hours and yields results that can be gamed by prompt engineering or benchmark contamination. We ask a different question: *can reasoning capacity be inferred directly from model weights, without running any inference?*

We focus on the three attention projection matrices W_Q , W_K , and W_V and their singular-value decompositions. Our central finding is that five universal geometric regularities appear across every model family we examine. The first three regularities (Laws I–III) concern the singular-value spectra of Q and K. Laws IV–V extend the picture to the geometric relationship among the *output subspaces* (left singular vectors, U -space) and the *input subspaces* (right singular vectors, V -space) of all three projections, revealing a complete static spectral architecture of Transformer attention.

B. Contributions

1. Formalization of **Wang’s Five Laws**: three empirical spectral regularities, one analytic precision-depth bound, and two universal orthogonality laws governing the geometric structure of attention subspaces.
2. Discovery of **super-orthogonality**: the output subspaces of V are systematically *below* the random baseline relative to Q and K, by approximately 20%, across all tested models and modalities.
3. Reproducible verification across five models spanning four architecture families (Qwen, DeepSeek, LLaMA, Gemma), including both text and vision modalities.
4. Quantitative evidence that RL fine-tuning systematically improves deep-layer spectral alignment (Second Law).
5. Practical guidance for training monitoring, checkpoint selection, and precision planning.

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II. BACKGROUND

A. Transformer Self-Attention

The scaled dot-product attention is defined as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right)V, \quad (1)$$

where $Q = XW_Q$, $K = XW_K$, $V = XW_V$, and d_h is the head dimension.

B. Singular Value Decomposition

For a weight matrix $M \in \mathbb{R}^{m \times n}$, the SVD is

$$M = U\Sigma V^\top, \quad \Sigma = \text{diag}(\sigma_1, \dots, \sigma_r), \quad \sigma_1 \geq \dots \geq \sigma_r \geq 0. \quad (2)$$

The columns of U are the *left singular vectors* (output subspace); the columns of V are the *right singular vectors* (input subspace). All spectral metrics in this paper are computed on W_Q , W_K , and W_V extracted from each attention head.

III. WANG’S FIVE LAWS

A. First Law — Spectral Linear Alignment

Statement. The Pearson correlation between the Q and K singular-value spectra converges toward a theoretical constant:

$$r(s_q, s_k) \rightarrow r_{\text{Wang}} = 1. \quad (3)$$

Wang’s Constant is defined as the theoretical extreme $r_{\text{Wang}} = 1$; in practice we observe 0.94–0.99.

Interpretation. High Pearson correlation indicates stable spectral alignment between query and key subspaces, supporting stable information propagation through deep layers.

Empirical evidence is summarized in Table I.

TABLE I. Median and mean Pearson correlation $r(s_q, s_k)$ and SSR across all layers and heads. Models are ordered by reasoning tier (weakest to strongest).

Model	Med. r	Mean r	Med. SSR	Mean SSR	Layers
Gemma-4-E2B	0.9183	0.9242	0.015702	0.013537	35
Gemma-4-E4B	0.9585	0.9411	0.009747	0.010080	42
LLaMA-3-8B	0.9813	0.9737	0.006196	0.007009	32
Qwen2.5-14B	0.9795	0.9710	0.006077	0.006710	48
DeepSeek-R1	0.9800	0.9714	0.005948	0.006585	48

B. Second Law — Spectral Shape Fidelity

Statement. The normalized spectral mismatch between Q and K,

$$\text{SSR} = \frac{1}{d_h} \sum_{i=1}^{d_h} |\tilde{s}_{q,i} - \tilde{s}_{k,i}|, \quad \tilde{s} = \frac{s}{\|s\|_2}, \quad (4)$$

decreases systematically in deep layers, with theoretical extreme $\text{SSR}_{\text{Wang}} = 0$.

RL amplifies alignment. Table II compares Qwen2.5-14B (base) and DeepSeek-R1 (RL-tuned) across four layer groups. The improvement grows monotonically with depth, reaching 3.92% in layers 36–47.

TABLE II. Layer-group mean SSR: Qwen2.5-14B vs. DeepSeek-R1.

Layer group	Qwen2.5-14B	DeepSeek-R1	Improvement
0–11	0.006852	0.006818	+0.48%
12–23	0.006414	0.006338	+1.17%
24–35	0.006831	0.006704	+1.87%
36–47	0.006743	0.006479	+3.92%

C. Third Law — Precision-Depth-Logic Criterion

Statement. The maximum trainable depth L_{max} is bounded by

$$L_{\text{max}} = \min(L_{\text{info}}, L_{\text{quant}}, L_{\text{dyn}}), \quad (5)$$

where

$$L_{\text{info}} = 1/\sqrt{\text{SSR}}, \quad (6)$$

$$L_{\text{quant}} = 3 \cdot 2^{2m} \quad (m = \text{mantissa bits}), \quad (7)$$

$$L_{\text{dyn}} = \log_{\kappa}(\text{MaxFinite}). \quad (8)$$

TABLE III. Dynamic-range depth limit L_{dyn} ($\kappa \approx 5$).

Format	m	MaxFinite	L_{dyn}
FP16	10	6.55×10^4	16
BF16	7	3.39×10^{38}	128

D. Fourth Law — Global Output-Subspace Decoupling

Statement. Let $U_Q, U_K, U_V \in \mathbb{R}^{d_h \times d_h}$ denote the matrices of left singular vectors of W_Q, W_K, W_V , respectively. Define the mean absolute cosine similarity of matched columns as

$$\overline{\cos}(U_A, U_B) = \frac{1}{d_h} \sum_{i=1}^{d_h} |\langle u_{A,i}, u_{B,i} \rangle|. \quad (9)$$

Then:

$$\overline{\cos}(U_Q, U_K) \sim \frac{1}{\sqrt{d_h}} \quad (\text{random orthogonality}), \quad (10)$$

$$\overline{\cos}(U_Q, U_V) < \frac{1}{\sqrt{d_h}} \quad (\text{super-orthogonality}), \quad (11)$$

$$\overline{\cos}(U_K, U_V) < \frac{1}{\sqrt{d_h}} \quad (\text{super-orthogonality}). \quad (12)$$

Interpretation. Equation (10) shows that the output subspaces of Q and K are mutually independent: they explore different directions in the representation space without systematic alignment. Equations (11)–(12) reveal that the output subspace of V is *super-orthogonal* to both Q and K: it actively avoids the directions used by the retrieval path, guaranteeing channel isolation between attention-score computation and content readout. We define *super-orthogonality* as a systematic deviation below the random expectation $1/\sqrt{d_h}$, typically by $\approx 20\%$.

Empirical evidence is given in Table IV. The pattern holds across all five models, including both text and vision modalities, and persists under RL fine-tuning.

E. Fifth Law — Global Random Orthogonality of Input Subspaces

Statement. Let $V_Q, V_K, V_V \in \mathbb{R}^{d_{\text{model}} \times d_h}$ denote the matrices of right singular vectors of W_Q, W_K, W_V , respectively. Define the mean absolute cosine similarity analogously to Eq. (9) but over the d_h matched right singular vectors. Then, compared with the random baseline $1/\sqrt{d_{\text{model}}}$:

$$\overline{\cos}(V_Q, V_K) \approx \overline{\cos}(V_Q, V_V) \approx \overline{\cos}(V_K, V_V) \approx \frac{1}{\sqrt{d_{\text{model}}}}. \quad (13)$$

Interpretation. The three projections select their sensitive directions from the full d_{model} -dimensional input embedding space independently and without a shared basis. This structural freedom allows each projection to specialize in complementary aspects of the input representation.

Empirical evidence is given in Table V. A slight elevation of $\overline{\cos}(V_Q, V_K)$ above baseline ($\sim 1.5\times$ in text models) represents an extremely weak coupling and does not constitute structural alignment. The vision encoder (Gemma-4-31B vision) shows a more pronounced elevation of $\overline{\cos}(V_Q, V_K)$ (0.044 vs. baseline 0.030), reflecting the requirement to share spatially sensitive directions in image self-attention; this does not alter the overall conclusion of near-random orthogonality.

Figure 1 provides a simultaneous layer-wise visualization of all metrics underlying Laws I, II, IV, and V for both models studied in this work.

F. Summary: A Complete Spectral Architecture

The five laws together describe the static spectral architecture of Transformer attention at the end of pre-training:

- **Σ -space** (singular values): full alignment between Q and K (First and Second Laws).
- **U-space** (output subspaces): Q–K random orthogonality plus V super-orthogonality, guaranteeing functional decoupling of retrieval and readout (Fourth Law).
- **V-space** (input subspaces): global random orthogonality, ensuring independent and free feature selection from the embedding space (Fifth Law).

This structure is highly consistent across multiple model families (Qwen, DeepSeek-R1, LLaMA-3, Gemma-4) and modalities (text and vision), indicating that it constitutes a *universal geometric fixed point* reached by pretraining, independent of specific architecture details or training objectives.

IV. METHODOLOGY

Models. We analyze five open-source models spanning four architecture families: LLaMA-3-8B [4], Qwen2.5-14B [5], DeepSeek-R1 [3], Gemma-4-E2B, and Gemma-4-E4B [6]. For Laws IV–V we additionally analyze Gemma-4-31B in both text and vision configurations.

Procedure.

1. Load weight tensors W_Q, W_K, W_V for each head and layer.
2. Compute SVD per Eq. (2), obtaining U, Σ, V for each matrix.
3. Compute Pearson $r(s_q, s_k)$ and SSR per Eqs. (3)–(4) (Laws I–III).
4. Compute $\overline{\cos}(U_A, U_B)$ and $\overline{\cos}(V_A, V_B)$ per Eq. (9) for all pairs $(A, B) \in \{(Q, K), (Q, V), (K, V)\}$ (Laws IV–V).
5. Compare against random baselines $1/\sqrt{d_h}$ (U-space) and $1/\sqrt{d_{\text{model}}}$ (V-space).
6. Aggregate across heads; record layer-wise and model-wide statistics.

No inference, tokenization, or prompt evaluation is required. All computation runs on CPU in under five minutes per model.

Verification scripts. `check-gemma.py`, `check-qwen.py`, `check-llama.py`, `check_r1_full.py`, `qwen-vs-deepseek-all-layers.py`, and `check-law4-law5.py` are available at <https://github.com/emis-framework/math-under-llm>.

TABLE IV. Mean absolute cosine similarity of matched left singular vectors (cross-model averages over all layers and heads). Bold entries are *below* the random baseline, indicating super-orthogonality.

Model	d_h	Baseline	$\overline{\cos}(U_Q, U_K)$	$\overline{\cos}(U_Q, U_V)$	$\overline{\cos}(U_K, U_V)$
Qwen2.5-14B-Instruct	128	0.0884	0.0981	0.0704	0.0702
DeepSeek-R1-Distill-Qwen-14B	128	0.0884	0.0982	0.0705	0.0699
LLaMA-3-8B	128	0.0884	0.0949	0.0707	0.0705
Gemma-4-31B (text)	256	0.0625	0.0630	0.0497	0.0500
Gemma-4-31B (vision)	128	0.0884	0.1024	0.0714	0.0713

TABLE V. Mean absolute cosine similarity of matched right singular vectors (cross-model averages over all layers and heads). All values are close to the random baseline $1/\sqrt{d_{\text{model}}}$.

Model	d_{model}	Baseline	$\overline{\cos}(V_Q, V_K)$	$\overline{\cos}(V_Q, V_V)$	$\overline{\cos}(V_K, V_V)$
Qwen2.5-14B-Instruct	5120	0.0140	0.0212	0.0142	0.0211
DeepSeek-R1-Distill-Qwen-14B	5120	0.0140	0.0211	0.0142	0.0210
LLaMA-3-8B	4096	0.0156	0.0258	0.0155	0.0234
Gemma-4-31B (text)	5376	0.0136	0.0167	0.0128	0.0153
Gemma-4-31B (vision)	1152	0.0295	0.0440	0.0306	0.0304

V. PRACTICAL APPLICATIONS

The five laws suggest several immediately actionable uses:

- **Benchmark-free reasoning assessment.** SSR and r can be computed from any checkpoint without inference, enabling fast model screening.
- **Checkpoint selection.** During training, the checkpoint with the lowest deep-layer SSR is predicted to yield the best reasoning performance.
- **RL progress monitoring.** SSR provides a layer-wise signal of RL-induced structural improvement, independent of reward hacking.
- **Precision planning.** The Third Law gives an analytic guide for choosing FP16 vs. BF16 based on target depth.
- **Architecture design.** The Fourth Law implies that any attention variant should preserve super-orthogonality between V and Q/K output subspaces; deviations may degrade reasoning fidelity.
- **SSR-regularized fine-tuning.** Adding $\lambda \sum_i |\text{SSR}_i^{\text{pre}} - \text{SSR}_i|$ to the fine-tuning loss anchors the spectral structure that encodes reasoning capability.

VI. FUTURE DIRECTIONS

1. **Validation on closed-source models.** We invite researchers with access to GPT-4/4.5, Claude 3, and Gemini weights to run the verification scripts and report anonymized SSR and cosine profiles.

2. **Theoretical foundations.** Why does gradient descent drive $r \rightarrow 1$ and induce super-orthogonality? Can these properties be derived from a single variational principle?
3. **Gauge-symmetry interpretation.** The spectral alignment between Q and K suggests an emergent $U(d_h)$ gauge symmetry in the attention mechanism. The super-orthogonality of V may correspond to flatness of the associated connection (Fourth Law as a flatness condition).
4. **Hilbert-space adjoint hypothesis.** If $W_Q \approx f(W_K^\dagger)$ for a spectral-preserving f , Q may be analytically derivable from K, bypassing gradient descent for attention layers.
5. **Spectral surgery.** Can reasoning capability be improved by directly modifying singular values of underperforming layers, without labeled training data?
6. **MoE and multi-modal extensions.** Do all five laws hold per expert in mixture-of-experts models? Does cross-modal attention exhibit different super-orthogonality behavior?
7. **Connection to Chern–Weil theory.** The U-space structure (Laws IV–V) may be related to characteristic classes of the principal bundle defined by the attention connection, providing a topological interpretation of the universal geometric fixed point.

VII. CONCLUSION

We have presented Wang’s Five Laws: a complete characterization of the static spectral architecture of Trans-

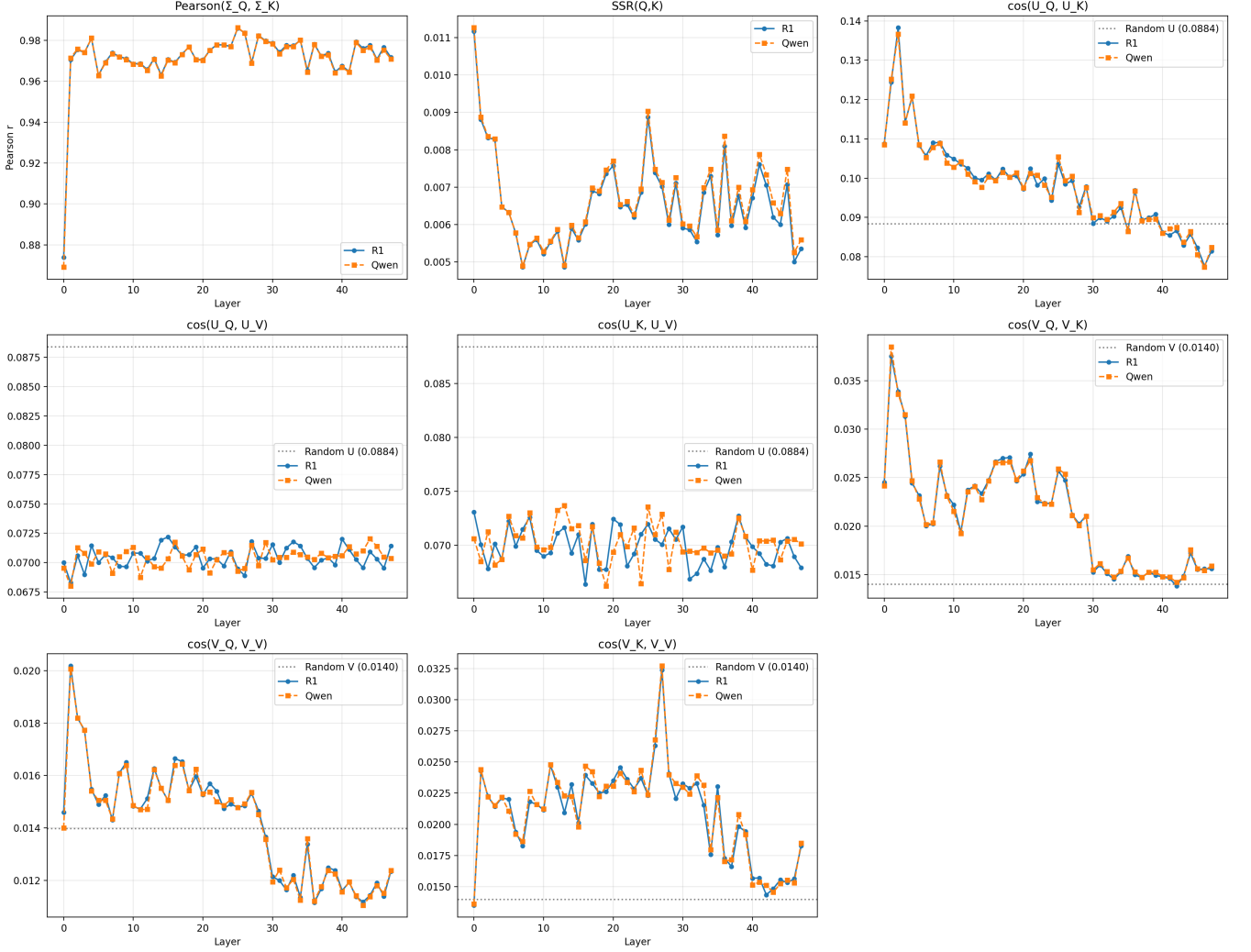


FIG. 1. Layer-wise spectral comparison between **Qwen2.5-14B-Instruct** (dashed red) and **DeepSeek-R1-Distill-Qwen-14B** (solid blue) across all 48 layers, averaged over all attention heads at each layer. Shaded bands indicate the interquartile range across heads. *Top row, left*: Pearson correlation $r(s_q, s_k)$ (First Law); the dotted horizontal line marks the empirical lower bound $r_{\text{Wang}} = 0.94$. *Top row, right*: Normalized spectral mismatch SSR (Second Law); deep-layer SSR is systematically lower in the RL-tuned model (DeepSeek-R1), with a gap reaching 3.92% in layers 36–47. *Middle row, left*: $\overline{\cos}(U_Q, U_K)$ remains near the random baseline $1/\sqrt{d_h} = 0.0884$ (Fourth Law, random-orthogonality branch). *Middle row, center and right*: $\overline{\cos}(U_Q, U_V)$ and $\overline{\cos}(U_K, U_V)$ lie *systematically below* the random baseline by $\approx 20\%$ across all layers (Fourth Law, super-orthogonality branch). *Bottom row*: $\overline{\cos}(V_Q, V_K)$, $\overline{\cos}(V_Q, V_V)$, and $\overline{\cos}(V_K, V_V)$ all remain close to the random baseline $1/\sqrt{d_{\text{model}}} = 0.0140$, confirming global random orthogonality of input subspaces (Fifth Law). The near-perfect overlap between the two models across all eight panels demonstrates that this spectral architecture is a universal fixed point of pretraining, fully preserved under RL fine-tuning.

former attention weights.

The First and Second Laws establish universal alignment of Q and K singular-value spectra, with RL fine-tuning deepening this alignment monotonically with layer depth. The Third Law provides an analytic ceiling on trainable depth as a function of floating-point precision. The Fourth Law reveals a novel *super-orthogonality* phenomenon: the output subspace of V systematically avoids the output subspaces of Q and K by approximately 20% below the random baseline, across all tested models and modalities. The Fifth Law confirms that input

subspaces of Q, K, V are globally random-orthogonal, ensuring independent feature selection.

Together, the five laws define a *universal geometric fixed point* of pretraining in the space of attention weight matrices: full spectral alignment in Σ -space, functional decoupling with super-orthogonality in U -space, and global random orthogonality in V -space. This structure is consistent across four architecture families and two modalities, suggesting it reflects a fundamental property of the attention mechanism rather than an artifact of any particular training recipe.

These findings enable *static, inference-free assessment of LLM reasoning capability* from attention weights alone, and open new theoretical questions about the geometric and topological foundations of large language models.

All results are reproducible from public model weights

using the accompanying scripts.

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